

Artificial Intelligence

Assignment 1

Agent Title: IT Helpdesk Router

Texture Form:

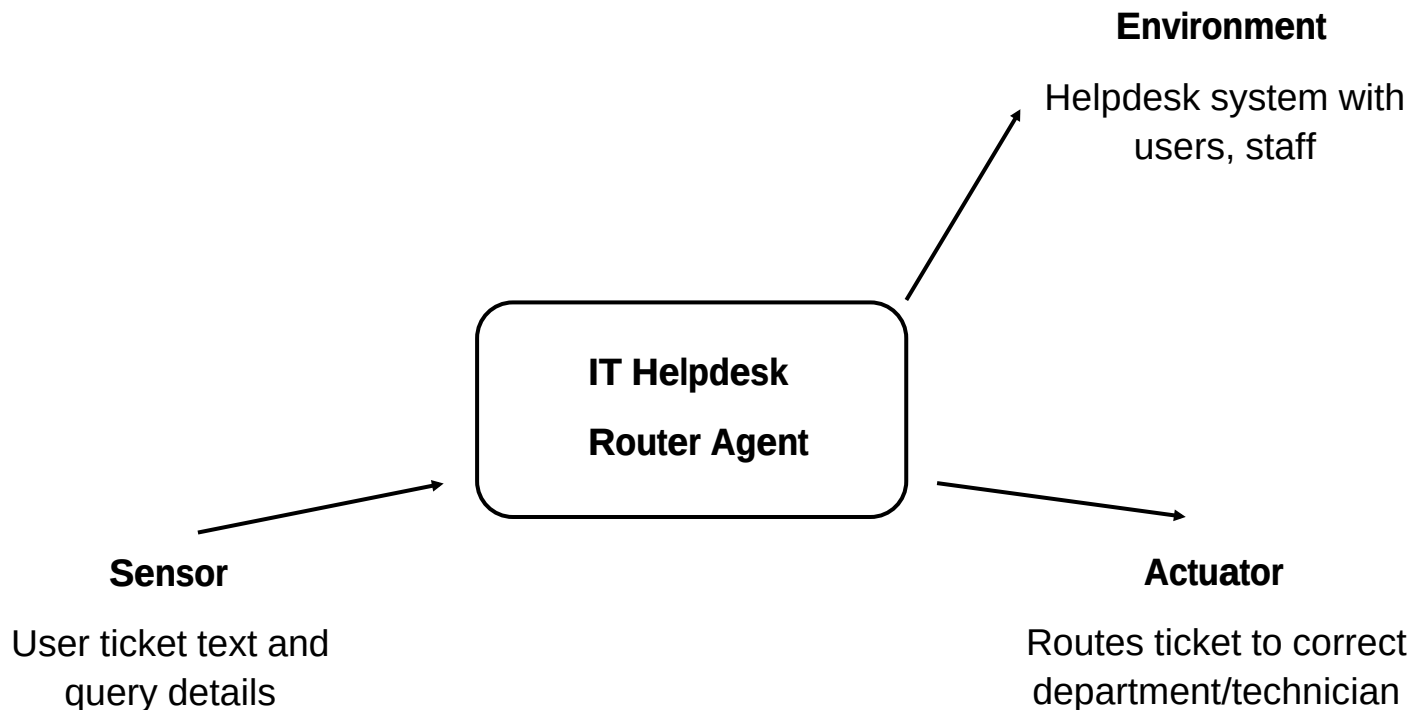
Sensor: Takes in the user's ticket/query (the text they typed, plus stuff like category or attachments if any).

Processing: Reads the ticket and figures out what it's about, then decides which department or person should handle it.

Actuator: Sends/assigns the ticket to the right department or technician and notifies them.

Formulation:

Graphical Representation:



PEAS Framework :

Performance → Fast, accurate ticket routing.

Environment → Helpdesk system, users, staff.

Actuator → Assigns ticket, notifies technician.

Sensor → User's ticket text/query.

Environment Type :

Property	Type	Why
Observability	Partially	agent can't see the full real picture
Determinism	Stochastic	same ticket could turn out differently
Episodes	Episodic	each ticket is handled on its own
Change	Dynamic	new issues and staff changes keep happening
Values	Discrete	routing options are fixed, limited categories
Agents	Multi	many users and staff are involved